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BIOMEDICAL LIMITED

WATER SYSTEM VALIDATION PROTOCOL

(WFI Production System – GMP Critical)

LOCATION – WATER TREATMENT PLANT
PROTOCOL No. – BML/QA/WSP/001

PROVIDED BY:
BIOMEDICAL LIMTED
KWARA STATE, NIGERIA

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1. PURPOSE AND SCOPE

1.1 Purpose

This Validation Protocol defines the strategy, methodology, acceptance criteria, and documentation requirements for the validation of the Water for Injection (WFI) production system. The validation demonstrates that the complete water system consistently produces WFI meeting the requirements of the European Pharmacopoeia (Ph. Eur.), British Pharmacopeia (BP), and applicable GMP regulations.

1.2 Scope

This protocol covers the entire water purification train from source water to WFI point-of-use, encompassing the following treatment stages:

- Borehole abstraction and raw water storage tanks
- Multimedia sand filtration
- Activated carbon filtration
- Water softening (ion exchange — cation resin, sodium regeneration)
- Ion exchange demineralisation (two-bed deionisation)
- Reverse Osmosis (RO) — double pass
- Multicolumn Distillation Plants (MCDP) for WFI generation
- WFI storage tanks (jacketed, with CIP capability)
- WFI distribution loop — all points of use

The following are excluded from this protocol:

- Process water (purified water) distribution loops not associated with the WFI system
- Steam generation systems validated under a separate steam qualification protocol
- Cleaning validation of downstream manufacturing equipment.

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2. REGULATORY REFERENCES AND STANDARDS

Reference	Description
Ph. Eur. 0169	Water for Injections — specification and production requirements
USP <1231>	Water for Pharmaceutical Purposes
EU GMP Annex 1 (2022)	Manufacture of Sterile Medicinal Products
EU GMP Chapter 3	Premises and Equipment
EU GMP Chapter 4	Documentation
ICH Q9	Quality Risk Management
ISPE Baseline Guide Vol. 4	Water and Steam Systems

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3. SYSTEM DESCRIPTION

3.1 Water Treatment Train Overview

The WFI production system is a multi-stage purification train designed to progressively reduce chemical, microbiological, and particulate contamination from borehole source water to pharmacopoeial WFI standard. The complete process train is described below:

3.1.1 Stage 1 — Borehole Abstraction & Raw Water Storage

Source water is abstracted from 2 boreholes equipped with submersible pumps. Raw water is transferred to buffer storage tanks (capacity: 10000 L) prior to treatment. Storage tanks are sealed with vent filters. Overflow and drain connections are indirect to prevent back-contamination.

3.1.2 Stage 2 — Multimedia Sand Filtration

Raw water passes through multimedia sand filter vessels containing graded layers of anthracite, sand, and gravel. This stage removes suspended solids, turbidity, and coarse particulates. Filters are backwashed automatically based on differential pressure and timed cycles. Backwash water is directed to drain.

3.1.3 Stage 3 — Activated Carbon Filtration

Granular activated carbon (GAC) filter vessels remove chlorine, chloramines, organic compounds, taste, and odour. Carbon beds are periodically sanitised by hydrogen peroxide solution to control microbial growth. The filter is also periodically backwashed to removed adsorbed matter. Bed replacement is based on performance monitoring and scheduled change-out intervals.

3.1.4 Stage 4 — Water Softening

Cation exchange resin vessels (sodium form) remove hardness ions (Ca^{2+}, Mg^{2+}) by exchange with sodium ions. Softeners are regenerated with pharmaceutical-grade sodium chloride solution on a timed-triggered basis. Hardness breakthrough is monitored and triggers manual regeneration.

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3.1.5 Stage 5 — Demineralised Water Production (Ion Exchange)

Demineralised water (DM water) is produced via two-bed ion exchange systems. Cation resin (H^+ form) removes cations; anion resin (OH^- form) removes anions. Conductivity and pH are continuously monitored. Exhausted resin beds are regenerated with acid (HCl) and caustic (NaOH). DM water is used as feed to the RO system.

3.1.6 Stage 6 — Reverse Osmosis (RO)

The RO system (double pass) uses semipermeable membranes operating at elevated pressure (typically 8–15 bar) to remove dissolved ions, organics, endotoxins, and microorganisms. Permeate conductivity, pressure, flow, and recovery rate are continuously monitored. Reject is directed to drain. Membranes are sanitised by peroxide water recirculation (periodic). RO permeate serves as feed water for the MCDP.

3.1.7 Stage 7 — Multicolumn Distillation Plant (MCDP)

The MCDP is the primary WFI generation unit. It operates on the principle of multiple-effect distillation, where feed water is evaporated and condensed across 4 columns (effects), each at progressively lower pressure and temperature. The first effect is heated by clean steam at 4 bar. Each subsequent effect is heated by the vapour generated in the preceding column. The WFI distillate is collected in a vapour separator to ensure separation from entrained water droplets and any residual pyrogens.

The MCDP operates continuously at temperatures ensuring bactericidal conditions (≥80°C for storage; distillation temperatures >121°C in early effects). This self-sanitising operation is fundamental to the hygienic design of the WFI system.

3.1.8 Stage 8 — WFI Storage and Distribution

Freshly distilled WFI is collected in jacketed storage tanks (capacity: 4000 L) maintained at ≥80°C (hot storage) to prevent microbial growth. Tanks are equipped with hydrophobic vent filters (0.22 µm), spray ball for CIP, level indication, temperature probes, and pressure relief. WFI is distributed continuously via a circulating loop (heated to ≥80°C) to all points of use (POU). All pipework is 316L stainless steel, electropolished internally, with orbital-welded connections.

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4. VALIDATION STRATEGY

4.1 Validation Life Cycle Approach

Validation follows the WHO -recommended three-phase approach for water systems, integrated within the overall validation life cycle:

Phase	Period	Description
Phase 1	2 weeks	Intensive monitoring during commissioning. All parameters, all sampling points, every day. Demonstrates initial compliance and identifies system characteristics.
Phase 2	2 weeks	Extended monitoring under normal operational conditions including seasonal variation. Sampling frequency remains high. Demonstrates consistent performance under routine operation.
Phase 3	12 months	Routine monitoring under full operational conditions across all seasons. Demonstrates long-term consistency and establishes alert/action limits. Transitions to ongoing monitoring program. Reduced testing frequency.

4.2 Qualification Activities

The following formal qualification documents are required and must be approved prior to the commencement of validation activities:

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Activity	Document Ref.	Scope
URS		User requirements for the complete water system
Risk Assessment		FMEA / risk assessment for water system components
DQ		Design Qualification of MCDP, RO, storage, and distribution
IQ	BML/ENGR/XXXIQX/001	Installation Qualification of all system components
OQ	BML/ENGR/XXXOQX/001	Operational Qualification — all instruments, alarms, controls
PQ Phase	BML/ENGR/XXXPQX/001	Performance Qualification Phase — intensive sampling

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5. CRITICAL QUALITY ATTRIBUTES (CQAs) AND ACCEPTANCE CRITERIA

5.1 WFI — Pharmacopoeial Specification

WFI must comply with Ph. Eur. 0169 and USP <1231> at all times. The following are the critical quality attributes:

Parameter	Ph. Eur. Limit	USP Limit	Alert Limit (Internal)
Conductivity (25°C)	≤1.3 µS/cm	≤1.3 µS/cm	≤0.5 µS/cm
TOC (Total Organic Carbon)	≤0.5 mg/L (500 ppb)	≤0.5 mg/L	≤0.25 mg/L
Endotoxins (LAL)	≤0.25 EU/mL	≤0.25 EU/mL	≤0.125 EU/mL
Bioburden (plate count)	≤10 CFU/100 mL	≤10 CFU/100 mL	≤5 CFU/100 mL
Nitrates	≤0.2 ppm	≤0.2 ppm	≤0.1 ppm
Heavy Metals	≤0.1 ppm	≤0.1 ppm	≤0.05 ppm
pH (informational)	5.0–7.0	5.0–7.0	5.5–6.5
Appearance	Clear, colourless	Clear, colourless	N/A
Particulates (≥10 µm)	Ph. Eur. 2.9.19	USP <788>	Per Ph. Eur. 2.9.19

**internal alert limit shall reset to statistical validated history limits trending.

5.2 Intermediate Water Quality Targets

The following in-process quality targets apply at each treatment stage to ensure adequate performance of upstream units and protection of downstream processes:

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Stage / Point	Conductivity	Bioburden	Other Key Parameter
Borehole / Raw Water	Report	Report	Turbidity ≤1 NTU target
Post Sand Filter	Report	≤500 CFU/mL	Turbidity ≤0.5 NTU
Post Carbon Filter	Report	≤100 CFU/mL	Chlorine = ND
Post Softener	Report	≤100 CFU/mL	Hardness ≤1 °dH
Post Ion Exchanger (DM)	≤5.1 µS/cm	≤50 CFU/mL	ROE, pH, TOC
Post RO Permeate	≤1.3 µS/cm	≤10 CFU/mL	ROE, pH, TOC
MCDP Feed Water	≤1.3 µS/cm	≤10 CFU/mL	Endotoxin ≤1 EU/mL, TOC, pH
WFI — MCDP Outlet	≤1.3 µS/cm	≤10 CFU/100 mL	Endotoxin ≤0.25 EU/mL, TOC, pH
WFI Storage Tank	≤1.3 µS/cm	≤10 CFU/100 mL	Temp ≥80°C maintained
WFI Loop — All POU	≤1.3 µS/cm	≤10 CFU/100 mL	Endotoxin ≤0.25 EU/mL

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6. CRITICAL PROCESS PARAMETERS (CPPs) AND MONITORING

6.1 MCDP Critical Process Parameters

Parameter	Normal Range	Alarm Limit	Action
Column 1 temperature	≥ 85 °C	< 85 °C	Stop production, investigate
Clean steam pressure (feed)	4 bar(g)	< 4 bar(g)	Alarm, check steam supply
Feed water flow rate	3000 L/hr	-	-
WFI conductivity (online)	≤1.3 µS/cm	>1.3 µS/cm alert	Divert to drain, investigate
Distillate temperature	≥ 85 °C	< 85 °C	Stop collection, divert to drain
Distillation rate / yield	400–3000 L/hr	Drop >20% from baseline	Inspect columns, check fouling

6.2 RO Critical Process Parameters

Parameter	Normal Range	Alarm Limit	Action
Feed pressure	100–150 bar	Outside range	Alarm, check pump/membranes
Permeate conductivity	≤5 µS/cm	>5 µS/cm	Alarm, check membrane integrity

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6.3 Storage Tank and Loop Critical Parameters

Parameter	Normal Range	Alarm Limit	Action
WFI tank temperature	≥80°C	<75°C	Alarm, do not use, heat up, investigate
Loop circulation flow	Continuous ≥ 3000 L/hr	Stopped	Restart immediately, investigate
Loop return temperature	≥80°C	<78°C	Alarm, increase heat trace/jacket
Tank level	10%–90%	Outside range	Alarm, adjust production/use

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7. SAMPLING PLAN

7.1 Phase 1 Sampling Plan (Weeks 1–2)

During Phase 1, intensive daily sampling is required at all sampling points throughout the entire water treatment train and distribution loop. No sampling point may be omitted during Phase 1.

Sampling Location	Frequency	Sample Volume	Tests Performed
PRE-TREATMENT			
Raw Water (Borehole)	3×/week	1000 mL	Conductivity, pH, hardness, turbidity, TCC, coliforms
Post Sand Filter	Daily	1000 mL	Turbidity, conductivity, bioburden
Post Carbon Filter	Daily	1000 mL	Conductivity, chlorine, bioburden, TOC
Post Softener	Daily	1000 mL	Hardness, conductivity, bioburden
DEMINERALISATION & RO			
Post Ion Exchanger (DM Water)	Daily	1000 mL	Conductivity, pH, ROE, bioburden, TOC, Nitrates,
RO Permeate	Daily	1000 mL	Conductivity, TOC, bioburden, endotoxin, pH
RO Reject	Weekly	1000 mL	Conductivity, TDS (for rejection calculation), TOC
WFI PRODUCTION — MCDP			

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Sampling Location	Frequency	Sample Volume	Tests Performed
MCDP Feed (RO product)	Daily	1000 mL	Conductivity, bioburden, endotoxin
MCDP Outlet / WFI Collection	Daily × 2 (AM/PM)	1000 mL	Full WFI spec: conductivity, TOC, endotoxin, bioburden, appearance, pH, nitrates, heavy metals
WFI STORAGE AND DISTRIBUTION			
WFI Storage Tank — Inlet	Daily	1000 mL	pH, Conductivity, TOC, endotoxin, bioburden
WFI Storage Tank — Outlet	Daily	1000 mL	Full WFI spec
WFI Loop — Return (pre-tank)	Daily	1000 mL	pH, Conductivity, TOC, endotoxin, bioburden
All POU Outlets (each)	Daily (rotate)	1000 mL	Full WFI spec — rotate so each POU tested ≥2×/week

7.2 Phase 2 Sampling Plan (Weeks 3–4)

Phase 2 sampling is reduced from Phase 1 but remains frequent. All WFI contact points must be sampled at least 3 times per week. Pre-treatment points may be sampled 3 times per week unless adverse trends are observed.

Sampling Zone	Frequency	Notes
Raw water / borehole	2×/week	Continue monitoring for seasonal changes

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Sampling Zone	Frequency	Notes
Pre-treatment (sand, carbon, softener)	2×/week	Monitor for microbial trends
Ion exchange DM water, RO permeate	3×/week	Include endotoxin 1×/week
MCDP outlet — WFI quality	Daily	Full spec; retain samples per SOP
WFI storage tanks	Daily	Monitor both inlet and outlet
WFI loop return	Daily	Conductivity online + grab sample 3×/week
All Points of Use	Minimum 3×/week each	Rotate on schedule; document rotation

7.3 Phase 3 / Routine Monitoring Sampling Plan

Phase 3 establishes the ongoing routine monitoring program. Routine monitoring continues indefinitely and data is reviewed at minimum monthly. Alert and action limits from Phases 1 and 2 are used to set statistical control limits.

Sampling Zone	Minimum Frequency	Notes
Borehole / raw water	2x / week	Seasonal variation assessment; full chemical panel annually
Pre-treatment stages	2×/week	Bioburden focus; conductivity

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Sampling Zone	Minimum Frequency	Notes
DM water / RO permeate	2×/week	pH, Conductivity, TOC, bioburden; endotoxin monthly
WFI — MCDP outlet	Daily	pH, Conductivity, TOC online; bioburden
WFI storage tanks	Daily	Temperature continuous; bioburden
WFI loop return	Daily	Conductivity continuous; bioburden
Points of Use	Weekly each	Full spec per POU; bioburden

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8. TEST METHODS AND ANALYTICAL PROCEDURES

8.1 Physicochemical Tests

Test	Method / Reference	Instrument	Frequency / Notes
Conductivity	Ph. Eur. 2.2.38 / USP <645>	Calibrated conductivity meter	Continuous online; daily grab
TOC	Ph. Eur. 2.2.44 / USP <643>	TOC analyser	Continuous or ≥daily; system suitability per Ph. Eur.
pH	Ph. Eur. 2.2.3	Calibrated pH meter (2-point cal)	Per sampling schedule
Nitrates	Ph. Eur. 2.4.20	UV spectrophotometry	Per sampling schedule
Heavy Metals	Ph. Eur. 2.4.8	AAS (outsourced or in-house method)	Quarterly or as per schedule
Hardness	APHA 2340C (EDTA titrimetric)	Titration or photometer	Post-softener, per schedule
Turbidity	ISO 7027 / NTU	Turbidimeter	Pre-treatment, per schedule
Chlorine (total/free)	DPD colorimetric / photometry	Photometer or titration	Post carbon filter, per schedule

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8.2 Microbiological Tests

Test	Method	Incubation / Reporting	Limit
Total Viable Count (TVC)	Membrane filtration (Ph. Eur. 2.6.12) on R2A or TGEA agar	30–35°C, 72 hr (R2A: 25°C, 5 days)	WFI: ≤10 CFU/100 mL
Endotoxins (LAL)	Gel clot or kinetic turbidimetric (Ph. Eur. 2.6.14 / USP <85>)	As per validated method; BET	WFI: ≤0.25 EU/mL
Total Coliform (raw water)	Membrane filtration on LTB/MLSB	35°C, 24–48 hr	Absent in 100 mL (raw water monitoring)
Microbial ID (OOS isolates)	MALDI-TOF or 16S rRNA sequencing	Per laboratory SOP	Identification of alert-triggering organisms
Environmental Monitoring	Settle plates, contact plates at POU areas	30–35°C, 48–72 hr	As per area classification limits

8.3 Instrument Calibration and System Suitability

- All instruments used for WFI testing are calibrated against traceable standards prior to each phase and at defined intervals (minimum annually)
- Online conductivity and TOC instruments undergo system suitability verification per Ph. Eur. 2.2.44 and 2.2.38 before and during each validation phase
- Endotoxin testing: BET validation (spike recovery, inhibition/enhancement testing) required per Ph. Eur. 2.6.14 for each water matrix tested

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- All reference standards are within expiry date and traceable to a national metrology institute (e.g., NIST, NPL, EURAMET)
- Calibration records are maintained and available for regulatory inspection

9. ALERT AND ACTION LIMITS — PROCEDURES

9.1 Definitions

Alert Limit: A level that, when exceeded, triggers investigation and increased monitoring but does not automatically require rejection of water or cessation of production. The alert limit is set below the action limit and below the pharmacopoeial specification.

Action Limit: A level that, when exceeded, requires immediate corrective action including cessation of supply from the affected point, investigation, and formal CAPA. Water produced during an exceedance may be quarantined pending investigation.

9.2 Alert and Action Limit Table — WFI

Parameter	Alert Limit	Action Limit	Specification Limit
Conductivity	≤0.4 µS/cm	>0.5 µS/cm	≤1.3 µS/cm
TOC	≤0.2 mg/L	>0.25 mg/L	≤0.5 mg/L
Endotoxins	≤0.125 EU/mL	≤0.25 EU/mL	≤0.25 EU/mL
Bioburden	≤4 CFU/100 mL	>7 CFU/100 mL	≤10 CFU/100 mL

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9.3 Alert Exceedance Response Procedure

1. Document the alert result in the QMS immediately
2. Notify QA and Engineering within the hour
3. Increase sampling frequency at the affected and adjacent points (double frequency for minimum 5 working days)
4. Review trending data for the past 30 days — look for upward trends or previous marginal results
5. Check CPP data (temperatures, pressures, flow rates) for the period surrounding the exceedance
6. Review maintenance, regeneration, and sanitisation logs for the affected system stage
7. Conduct root cause investigation — complete within 2 working days
8. Complete an Investigation Report and close with documented conclusion
9. Return to normal sampling frequency only after ≥5 consecutive compliant results at the affected point

9.4 Action Limit Exceedance Response Procedure

10. Immediately quarantine any water produced from the affected storage tank or loop zone
11. Cease supply to manufacturing processes — notify Production Manager and QA Responsible Person
12. Isolate the affected section if possible without compromising the entire loop
13. Open a formal deviation / non-conformance in the QMS
14. Conduct emergency sampling at all upstream stages to identify the source
15. Perform system flush and sanitisation per Sanitisation SOP
16. Sample post-sanitisation (minimum 3 consecutive samples per affected point) — must meet specification before resuming supply
17. Complete full CAPA including effectiveness check
18. Brief senior management and consider regulatory notification if systemic failure is indicated
19. Review and update risk assessment if root cause reveals new failure mode

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10. SANITISATION AND MAINTENANCE REQUIREMENTS

10.1 WFI Storage and Distribution — Hot Water Sanitisation

The primary sanitisation mechanism for the WFI storage and distribution system is maintenance of ≥80°C at all times during hot operation. In addition, periodic thermal sanitisation cycles are performed at ≥121°C (where system design permits) or ≥80°C for extended periods per approved SOP.

Component	Sanitisation Method	Frequency / Criteria
WFI storage tanks	Hot water CIP (≥80°C)	Minimum weekly formal CIP; continuous hot operation
WFI distribution loop	Hot water circulation (≥80°C)	Continuous — temperature monitored at return
MCDP	Self-sanitising (operating temperature)	Continuous — operating temp ≥80°C; full drain/flush quarterly
RO membranes	chemical (H_2O_2)	Weekly water flush; chemical per SOP if indicated by bioburden trend and then monthly chemical flush
Carbon filters	chemical sanitisation	Monthly minimum; or when bioburden trend increases
Ion exchange resin beds	chemical flush (H_2O_2)	Following regeneration cycle; full sanitise Monthly chemical flush.

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Component	Sanitisation Method	Frequency / Criteria
Sand filters	Backwash + chemical flush	formal wash monthly
Tank vent filters	Integrity test; replace	Test every 6 months; replace annually or on failure
Sampling valves / POU valves	Flaming before sampling	Each sampling event for microbiological testing, per sampling SOP

10.2 Planned Maintenance Impact on Validation

All planned maintenance activities on the water system must be assessed for potential impact on validated state. The following rules apply:

- Any replacement of membrane, resin, or filter media requires post-change sampling before WFI production resumes — minimum 3 consecutive compliant samples at all downstream points
- Any break in the WFI loop (opening of sealed pipework, replacement of gaskets or valves) requires sanitisation and re-qualification sampling — minimum 5 consecutive compliant samples
- MCDP column inspections require full recommissioning and at minimum 2 days of intensive sampling before resumption of normal production
- Changes to CPP setpoints (temperatures, pressures) require formal change control and OQ re-testing
- All maintenance must be documented in the Maintenance Management System (MMS) and cross-referenced to the water system logbook

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11. DEVIATION MANAGEMENT, OOS RESULTS, AND CAPA

11.1 Out-of-Specification (OOS) Result Handling

An OOS result is any analytical result that fails to meet the pharmacopoeial specification or the internal action limit for a defined parameter. The following procedure must be followed for all OOS results:

20. Phase 1 — Laboratory Investigation: Review the original analytical data, instrument calibration records, standard preparation, and analyst technique for assignable cause. Complete within 1 working day.
21. Phase 2 — Invalidation (if laboratory error confirmed): If a confirmed laboratory error is identified (instrument failure, transcription error, reagent OOS), the original result may be invalidated. Full documentation is required. Re-test the retained sample or original sample volume.
22. Phase 3 — System Investigation (if no laboratory error): If no laboratory cause is found, commence full system investigation. Sample all upstream stages. Review CPP data. Engage Engineering and QA.
23. Phase 4 — Corrective Action: Implement corrective action based on root cause (e.g., sanitise, replace component, adjust setpoint, increase regeneration frequency).
24. Phase 5 — Verification Sampling: Following corrective action, collect a minimum of 3–5 consecutive compliant results from the affected point before returning to routine monitoring.
25. Phase 6 — CAPA and Close-Out: Complete formal CAPA document. Assign effectiveness check with defined timeframe (minimum 30 days post-implementation). Close CAPA with QA approval.

11.2 Validation Failure Criteria

The validation is considered to have failed if any of the following occur during Phases 1 or 2:

- **Any WFI result fails the pharmacopoeial specification (Ph. Eur. 0169 or USP <1231>) without a confirmed laboratory invalidation**

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- **More than two alert limit exceedances at the same sampling point within any 10-day period**
- **Any confirmed endotoxin result >0.25 EU/mL in WFI**
- **Any confirmed bioburden result >10 CFU/100 mL in WFI**
- **Failure to maintain critical process parameter controls for a period exceeding [X] hours without immediate recovery**

In the event of validation failure, Phase 1 must be repeated from the beginning after the root cause has been identified, corrective action implemented, and system re-qualified.

12. DOCUMENTATION AND DATA MANAGEMENT

12.1 Required Validation Documents

Document	Owner	Requirement
This Validation Protocol (VP)	QA	Approved before Phase 1 start
Validation Report — Phase 1	QA	Completed within 10 WD of Phase 1 end
Validation Report — Phase 2	QA	Completed within 10 WD of Phase 2 end
Validation Report — Phase 3 (Annual)	QA	Annual review and report
Sampling Schedules / SOPs	QC Laboratory	Approved before sampling begins
Analytical Methods / SOPs	QC Laboratory	Validated / verified before use
Instrument Calibration Records	QC Laboratory	Current at time of use

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Document	Owner	Requirement
Water System Logbook	Engineering / Production	Real-time; all events, maintenance, deviations
CPP Continuous Data (	Engineering	Archived; minimum 5 years retention
LIMS /Laboratory Data Records	QC	21 CFR Part 11 / EU Annex 11 compliant
Deviation / OOS Records	QA	Linked to relevant sampling events
CAPA Records	QA	Closed with effectiveness verification
Change Control Records	QA / Engineering	All changes post-validation must be controlled

12.2 Data Integrity Requirements

- All raw data (analytical results, instrument printouts, meta data) must be attributable, legible, contemporaneous, original, and accurate (ALCOA+ principles)
- Electronic data must comply with Annex 11 (EU) and/or 21 CFR Part 11 (FDA) as applicable
- Handwritten records must be signed, dated, and written in indelible ink; corrections made with single strikethrough, signed, and dated
- LIMS records if available are the master record; LIMS must be validated and access controlled
- Audit trails for all electronic systems must be enabled, reviewed regularly, and retained for a minimum of 5 years (or product lifetime + 1 year if longer)
- Sample chain of custody must be documented from collection to result entry

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13. CHANGE CONTROL AND RE-QUALIFICATION

13.1 Changes Requiring Re-qualification

The following changes to the water system must be subject to formal change control and assessed for re-qualification impact before implementation:

Change Type	Minimum Re-qualification	Notes
Replacement of MCDP column(s)	OQ + PQ Phase 1 repeat	Full IQ if new equipment
RO membrane replacement (all or partial)	OQ re-test + 2 weeks intensive sampling	Salt rejection test required
New borehole or change in raw water source	PQ Phase 1 full repeat	Full chemical characterisation of new source
Addition of new POU point to loop	Sampling at new POU for 4 weeks before routine	Loop flow modelling review
Change in WFI storage temperature setpoint	OQ re-test; PQ Phase 2 repeat	Risk assessment required
Change in sanitisation agent/method	Method validation + PQ Phase 1 repeat	Microbial efficacy data required
Major pipework modification (loop)	IQ update + OQ + 4 weeks intensive sampling	P&ID update required
Carbon bed media replacement	OQ performance test + 2 weeks monitoring	Chlorine removal efficacy

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Change Type	Minimum Re-qualification	Notes
Ion exchange resin replacement	OQ + conductivity verification + 2 weeks monitoring	New resin qualification data
Software / SCADA changes	Computer system validation (CSV) review	Annex 11 / GAMP5 compliant

14. ROLES AND RESPONSIBILITIES

Role	Responsibilities
Factory Manager/Supt Pharm	Protocol approval; final sign-off on validation reports; regulatory accountability; approval of deviations with quality impact
QA Manager	Review and approval of protocol and reports; deviation review and CAPA oversight; change control; data integrity governance; preparation of regulatory submissions
QC Chemist	Protocol authorship; coordination of validation activities; compilation of reports; facilitation of investigations; management of validation master plan entry
QC Head	Approval of sampling SOPs and analytical methods; oversight of sampling execution; LIMS management; OOS investigations; laboratory calibration programme

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Role	Responsibilities
QC Analysts	Execution of sampling per approved SOPs; preparation and analysis of samples; recording of results in LIMS; flagging of OOS results to QC Manager
Engineering Manager	IQ/OQ execution and approval; maintenance programme management; CPP data provision; technical investigation support; change control technical review
Production Manager	Scheduling of validation activities; coordination with manufacturing; reporting of process deviations with water system impact; operator training oversight
Process Operator / Process Engineer	Daily operation of water system per SOPs; completion of water system logbook; reporting of abnormal conditions; execution of sanitisation procedures

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15. GLOSSARY OF ABBREVIATIONS

Abbreviation	Definition
ALCOA+	Attributable, Legible, Contemporaneous, Original, Accurate (+ Complete, Consistent, Enduring, Available)
BET	Bacterial Endotoxins Test
BMS	Building Management System
CAPA	Corrective and Preventive Action
CFR	Code of Federal Regulations
CFU	Colony Forming Units
CIP	Clean In Place
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CSV	Computer System Validation
DM Water	Demineralised Water
DQ	Design Qualification
EU	Endotoxin Units
EU GMP	European Union Good Manufacturing Practice
FMEA	Failure Mode and Effects Analysis
GAC	Granular Activated Carbon
GMP	Good Manufacturing Practice

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Abbreviation	Definition
IQ	Installation Qualification
ISPE	International Society for Pharmaceutical Engineering
LAL	Limulus Amebocyte Lysate
LIMS	Laboratory Information Management System
MCDP	Multicolumn Distillation Plant
MMS	Maintenance Management System
NTU	Nephelometric Turbidity Units
OQ	Operational Qualification
OOS	Out of Specification
P&ID	Piping and Instrumentation Diagram
Ph. Eur.	European Pharmacopoeia
POU	Point of Use
PQ	Performance Qualification
QA	Quality Assurance
QC	Quality Control
QMS	Quality Management System
QP	Qualified Person
rFC	Recombinant Factor C (endotoxin test)
RO	Reverse Osmosis
SCADA	Supervisory Control and Data Acquisition

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Abbreviation	Definition
SOP	Standard Operating Procedure
TOC	Total Organic Carbon
TVC	Total Viable Count
URS	User Requirements Specification
USP	United States Pharmacopeia
WFI	Water for Injections

16. APPENDICES

Appendix A — Sampling Location Map / P&ID Reference

The sampling locations defined in this protocol are identified on the approved P&ID drawing(s):

- Drawing No.: [P&ID-WS-001] — Water Treatment Train (Borehole to RO)
- Drawing No.: [P&ID-WS-002] — MCDP and WFI Storage/Distribution
- Sampling points are marked with tag numbers SP-WS-001 through SP-WS-[XX]

Note: A current, approved P&ID must be available in the validation report. Any P&ID revisions must be captured under change control.

Appendix B — Sampling Record Template

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Sample ID	Date/Time	Location / POU	Sample Vol (mL)	Analyst	Tests Requested

Appendix C — WFI System Sanitisation Log Template

Date	Component Sanitised	Method Used	Duration	Operator	QA Review

Appendix D — Phase Summary Acceptance Checklist

The following checklist must be completed and signed by QA prior to approval of each validation phase:

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Checklist Item	Yes / No	Initials
All sampling events completed per approved schedule		
No unresolved OOS results at close of phase		
All deviations have been documented and closed (or justified open)		
All CPP data has been reviewed and trends documented		
All instrument calibrations were current during sampling period		
Water system logbook is complete with no unexplained gaps		
All acceptance criteria have been met for this phase		
Phase validation report has been reviewed and is ready for approval		

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